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(54) **CRYSTALS OF DIAZABICYCLOOCTANE DERIVATIVE AND PRODUCTION METHOD FOR CRYSTALS OF DIAZABICYCLOOCTANE DERIVATIVE**

KRISTALLE EINES DIAZABICYCLOOCTANDERIVATS UND HERSTELLUNGSVERFAHREN FÜR KRISTALLE EINES DIAZABICYCLOOCTANDERIVATS

CRISTAUX DE DÉRIVÉ DE DIAZABICYCLOOCTANE ET PROCÉDÉ DE PRODUCTION DE CRISTAUX DE DÉRIVÉ DE DIAZABICYCLOOCTANE

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EP 3 067 355 B1

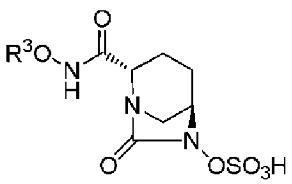
## Description

TECHNICAL FIELD

**[0001]** The present invention relates to a process for producing crystalline forms of a diazabicyclooctane derivative of formula (VII), particularly Formula (VII-1).

BACKGROUND ART

**[0002]** JP-B-4515704 indicates a novel heterocyclic compound, a production process thereof, and the use thereof as a pharmaceutical agent, and discloses as an example of a typical compound thereof sodium trans-7-oxo-6-(sulfooxy)-1,6-diazabicyclo [3.2.1]octane-2-carboxamide (NXL104). Production processes of a specific piperidine derivative as an intermediate are also indicated in JP-A-2010-138206 and JP-A-2010-539147, while a production process of NXL104 and crystalline forms thereof is disclosed in WO 2011/042560.
**[0003]** In addition, JP-B-5038509 indicates (2S,5R)-7-oxo-N-(piperidin-4-yl)-6-(sulfooxy)-1,6-diazabicyclo[3.2.1]octane-2-carboxami de (MK7655), while a production process of a specific piperidine derivative and MK7655 is disclosed in JP-A-2011-207900 and WO 2010/126820.
**[0004]** The inventors of the present invention also disclosed a novel diazabicyclooctane derivative of Formula (VII) in JP-A-2012-122603 (wherein $R^3$ is same as will be subsequently described):



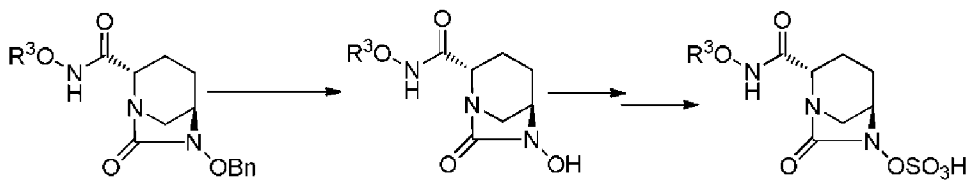
**VII**
**VII-1**: $R^3$=-$CH_2CH_2NH_2$

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

**[0005]**

Scheme **1**



**IV** **IV-1**: $R^3$=-$CH_2CH_2NH$-Boc; **V** **V-1**: $R^3$=-$CH_2CH_2NH$-Boc; **VII** **VII-1**: $R^3$=-$CH_2CH_2NH_2$

In the above Formulas, $R^3$ is same as will be subsequently described, OBn is benzyloxy and Boc is tert-butoxycarbonyl.
**[0006]** During the course of examining industrialization of (2S,5R)-N-(2-aminoethoxy)-7-oxo-6-(sulfooxy)-1,6-diazabicyclo[3.2.1]octane-2-carboxamide of the aforementioned Formula (VII), particularly Formula (VII-1), various problems relating to production were indicated, such as 1) the need to provide the compound in a stable crystalline form due to the handling difficulty during production of the amorphous active pharmaceutical ingredient (API), particularly the lyophilized product, and the difficulty in ensuring stability; 2) the need to isolate the crude active pharmaceutical ingredient containing an acid obtained after deprotecting the protecting group in the side chain $R^3$ONHC(=O)- group and to establish a procedure for adjusting to a stable pH range; 3) the need to improve yield and to avoid contamination with byproduct by controlling overreaction that causes the side chain $R^3$ONHC(=O)- group to also be sulfated during sulfation of a compound of Formula (V); 4) being unable to ignore isolation loss in addition to instability in solution state, particularly instability during evaporation of the reaction solvent to concentrate the compound of the aforementioned Formula (V);

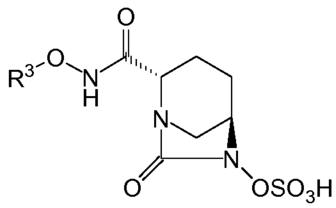
and 5) unsatisfactory yield of the compound of the aforementioned Formula (IV). In particular, the step from isolation and pH adjustment of the crude compound of Formula (VII-1-CR) containing an acid to crystallization of the compound of Formula (VII-1) was extremely difficult due to complex factors in the effects of instability and high solubility of compound, decomposition products and contaminants.

Means for Solving the Problems

**[0007]** The inventors conducted detailed studies on a process for producing the compound of the aforementioned Formula (VII) and established a series of production processes for providing a highly pure solution of the compound of Formula (VII) that does not affect crystallization of the compound of Formula (VII), as well as a process for producing highly stable crystalline forms.

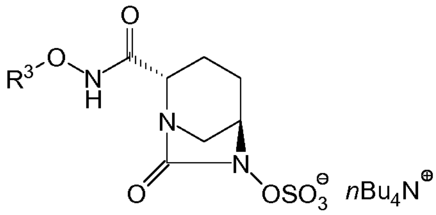
**[0008]** Namely, the present invention relates to

(1) a process for producing a compound of Formula (VII):



(VII)

wherein $R^3$ is $C_{1-6}$-alkyl or heterocyclyl, $R^3$ may be modified with 0-5 groups $R^4$ selected from $C_{1-6}$-alkyl, heterocyclyl, $R^5(R^6)N$- and a protecting group; and $R^4$ may be consecutively substituted ; $R^5$ and $R^6$ each independently is H or $C_{1-6}$ alkyl or together form heterocyclyl; and further, $R^3$, $R^5$ and $R^6$ can undergo ring closure at an arbitrary position; from a compound of formula (VI)



(VI)

comprising

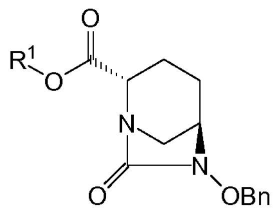
(iii) if the $R^3ONHC(=O)$- side chain has a protecting group, removing the protecting group with an acid, followed by (iv) precipitating a crude product, by adding a poor solvent selected from ethyl acetate, isopropyl acetate and butyl acetate to the reaction solution, to obtain a crude compound of formula (VII).

**[0009]** In preferred embodiments the invention provides the following:

(2) The process as defined in (1), further comprising the subsequent step of
(v) alternately adding the crude compound of formula (VII) and ice-cold buffer to obtain a solution having a pH of 4-5.5, concentrating after desalting with a synthetic adsorbent as necessary, adjusting the temperature, seeding as necessary and crystallizing by adding a poor solvent selected from methanol, ethanol and isopropanol to produce the compound of formula (VII).

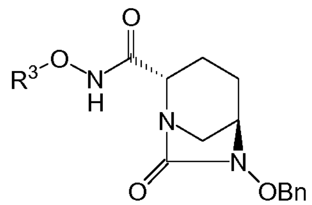
(3) The process as defined in (1) or (2), further comprising, prior to steps (iii) and (iv), the steps of

(i) reacting a compound of formula (III):



(III)

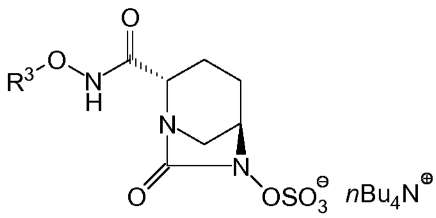
Wherein OBn is benzyloxy, and $R^1$ is 2,5-dioxopyrrolidin-1-yl, 1,3-dioxo-3a,4,7,7a-tetrahydro-1H-isoindol-2(3H)-yl, 1,3-dioxohexahydro-1H-isoindol-2(3H)-yl, or 3,5-dioxo-4-azatricyclo[5.2.1.$0^{2.6}$]dec-8-en-4-yl, with a compound $R^3ONH_2$ wherein $R^3$ is as defined in claim 1, to obtain a compound of formula (IV):



(IV);

and

(ii) treating with a palladium carbon catalyst under a hydrogen atmosphere, simultaneously or consecutively subjecting to a sulfation reaction using sulfur trioxide-trimethylamine complex in the presence of a catalytic amount of base in a hydrous solvent and treating with tetrabutylammonium hydrogensulfate to obtain a compound of formula (VI):

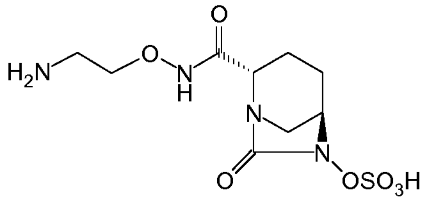


(VI).

(4) The process as defined in any of (1)-(3), wherein $R^3$ is selected from:

2-(tert-butoxycarbonylamino)ethyl; 2-aminoethyl;
2-((tert-butoxycarbonyl)(methyl)amino)ethyl; 2-(methylamino)ethyl;
2-((tert-butoxycarbonyl)(isopropyl)amino)ethyl; 2-(isopropylamino)ethyl;
2-(dimethylamino)ethyl; (2S)-2-((tert-butoxycarbonyl)amino)propyl;
(2S)-2-(amino)propyl; (2R)-2-((tert-butoxycarbonyl)amino)propyl; (2R)-2-(amino)propyl;
3-((tert-butoxycarbonyl)amino)propyl; 3-(amino)propyl;
(2S)-tert-butoxycarbonylazetidin-2-ylmethyl; (2S)-azetidin-2-ylmethyl;
(2R)-tert-butoxycarbonylpyrrolidin-2-ylmethyl; (2R)-pyrrolidin-2-ylmethyl;
(3R)-tert-butoxycarbonylpiperidin-3-ylmethyl; (3R)-piperidin-3-ylmethyl;
(3S)-tert-butoxycarbonylpyrrolidin-3-yl; (3S)-pyrrolidin-3-yl;
1-(tert-butoxycarbonyl)azetidin-3-yl; and azetidin-3-yl.

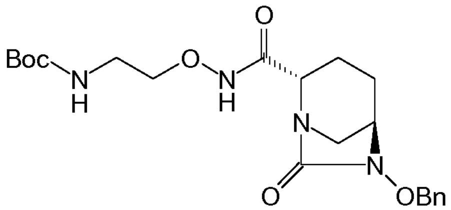
(5) The process as defined in (3), wherein the compound of formula (VII) is a compound of formula (VII-1)



(VII-1)

and wherein

- in step (i) the compound of formula (III) is reacted with tert-butyl 2-(aminooxy)ethylcarbamate in the presence of a base to obtain a compound of formula (IV-1) wherein Boc is tert-butoxycarbonyl:



(IV-1)

- in step (iii) the Boc protecting group is removed with trifluoroacetic acetic acid,
- in step (iv) ethyl acetate is used as the poor solvent, and
- in step (v) the ice-cold buffer is an ice-cold phosphate buffer, and isopropanol is used as the poor solvent.

(6) The process as defined in (1), wherein $R^3$ is 2-(tert-butoxycarbonylamino)ethyl, and

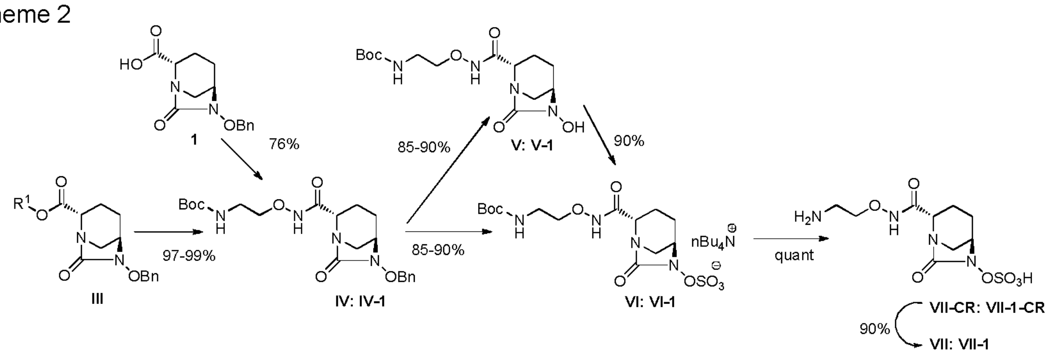
- in step (iii) the tert-butoxycarbonyl (Boc) protecting group is removed with trifluoroacetic acid, and
- in step (iv) the poor solvent is ethyl acetate.

(7) The process as defined in (2), wherein $R^3$ is 2-(amino)ethyl, the ice-cold buffer is an ice-cold phosphate buffer and isopropanol is used as the poor solvent.

Effects of the Invention

**[0010]** According to the series of production processes of the present invention, crystalline forms of a compound of formula (VII), particularly a compound of formula (VII-1) and crystalline forms thereof having favorable stability, can be produced with good reproducibility and high yield.

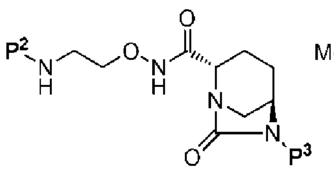
Scheme 2



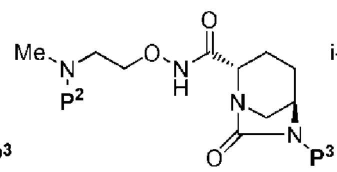

**[0011]** Although there were factors that directly impaired crystallization of the compound of formula (VII) and that was caused from decomposition of the compound of formula (VII) during adjustment of pH and from carrying over of degraded products in a series of steps to the next step, decomposition of the compound of formula (VII) is completely controlled, and a solution with high purity of the compound of formula (VII) that is able to be crystallized can be obtained at high yield by employing processes for producing a compound of formula (VI) and a compound of formula (VII-CR) established

tetrabutylammonium tert-butyl (2S)-2-{[({[(2S,5R)-7-oxo-6-(sulfooxy)-1,6-diazabicyclo[3.2.1]oct-2-yl]carbonyl}amino)oxy]methyl}azetidine-1-carboxylate;
(2S,5R)-N-[(2S)-azetidin-2-ylmethoxy]-7-oxo-6-(sulfooxy)-1,6-diazabicyclo[3.2. 1]octane-2-carboxamide;
tert-butyl (2R)-2-{[([(2S,5R)-6-benzyloxy-7-oxo-1,6-diazabicyclo[3.2.1] oct-2-yl]carbonyl}amino)oxy]methyl}-carboxylate;
tetrabutylammonium tert-butyl (2R)-2-{[({[(2S,5R)-7-oxo-6-(sulfooxy)-1,6-diazabicyclo [3.2.1]oct-2-yl]carbonyl}amino)oxy]methyl}pyrrolidine-1-carboxylate;
(2S,5R)-7-oxo-N-[(2R)-pyrrolidin-2-ylmethoxy]-6-(sulfooxy)-1,6-diazabicyclo[3. 2.1]octane-2-carboxamide;
tert-butyl (3R)-3-{[({[(2S,5R)-6-benzyloxy-7-oxo-1,6-diazabicyclo[3.2.1]oct-2-yl]carbonyl}amino)oxy]methyl}piperidine-1-carboxylate;
tetrabutylammonium tert-butyl (3R)-3-{[({[(2S,5R)-7-oxo-6-(sulfooxy)-1,6-diazabicyclo [3.2.1]oct-2-yl]carbonyl}amino)oxy]methyl}piperidine-1-carboxylate;
(2S,5R)-7-oxo-N-[(3R)-piperidin-3-ylmethoxy]-6-(sulfooxy)-1,6-diazabicyclo[3.2 .1]octane-2-carboxamide;
tert-butyl (3S)-3-[({[(2S,5R)-6-benzyloxy-7-oxo-1,6-diazabicyclo[3.2.1]oct-2-yl]carbonyl}amino)oxy]pyrrolidine-1-carboxylate;
tetrabutylammonium tert-butyl (3S)-3-[({[(2S,5R)-7-oxo-6-(sulfooxy)-1,6-diazabicyclo[3.2.1]oct-2-yl]carbonyl}amino)oxy]pyrrolidine-1-carboxylate;
(2S,5R)-7-oxo-N-[(3S)-pyrrolidin-3-yloxy]-6-(sulfooxy)-1,6-diazabicyclo[3.2.1]o ctane-2-carboxamide;
tert-butyl 3- {[({[(2S,5R)-6-benzyloxy-7-oxo-1,6-diazabicyclo[3.2.1]oct-2-yl] carbonyl}amino)oxy]methyl}azetidine-1-carboxylate;
tetrabutylammonium tert-butyl 3-{[({[(2S,5R)-7-oxo-6-(sulfooxy)-1,6-diazabicyclo[3.2.1]oct-2-yl]carbonyl}amino)oxy]methyl}azetidine-1-carboxylate; and
(2S,5R)-N-(azetidin-3-ylmethoxy)-7-oxo-6-(suliboxy)-1,6-diazabicyclo[3.2.1]oct ane-2-carboxamide,

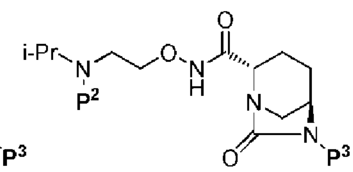
and include the following group of compounds:



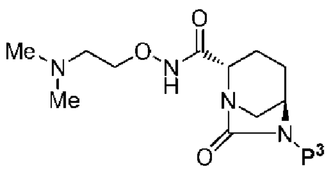
**IV-1, VI-1, VII-1**



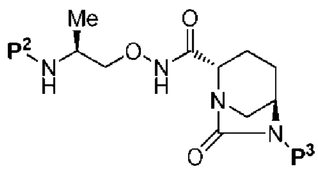
**IV-2, VI-2, VII-2**



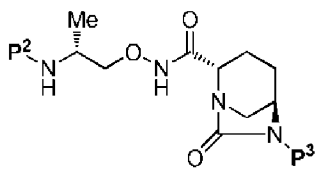
**IV-3, VI-3, VII-3**



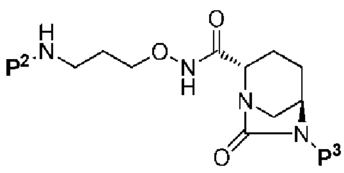
**IV-4, VI-4, VII-4**



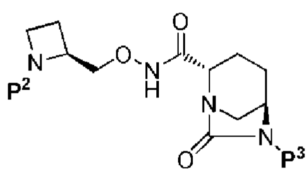
**IV-5, VI-5, VII-5**



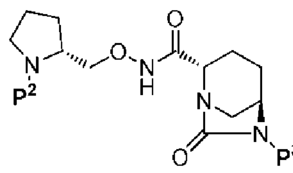
**IV-6, VI-6, VII-6**



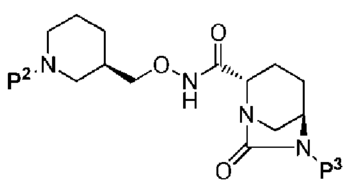
**IV-7, VI-7, VII-7**



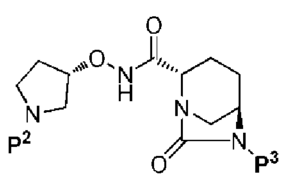
**IV-8, VI-8, VII-8**



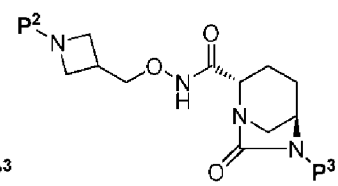
**IV-9, VI-9, VII-9**



**IV-10, VI-10, VII-10**



**IV-11, VI-11, VII-11**

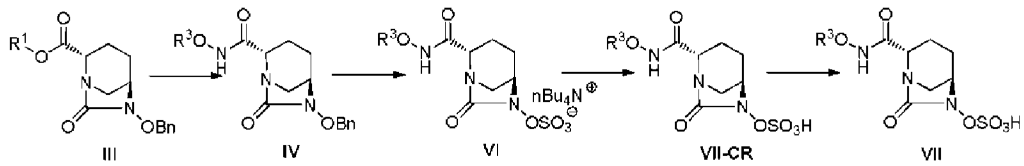


**IV-12, VI-12, VII-12**

preferable for concomitant administration with a compound of formula (VII) include imipenem, meropenem, biapenem, doripenem and ertapenem.

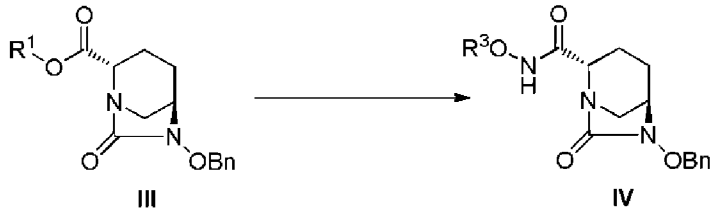
**[0051]** An example of a β-lactam antibiotic other than carbepenems, penicillins and cephems antibiotics that is particularly preferable for concomitant administration with a compound of formula (VII) is aztreonam.

**[0052]** The following provides sequential explanations of processes for producing a compound of formula (VII) and crystals thereof provided by the present invention.



In the above Formulas (III), (IV), (VI), (VII-CR) and (VII), OBn, $R^1$ and $R^3$ are same as described above.

**[0053]** The step for obtaining a compound of formula (IV) from a compound of Formula (III):



(wherein OBn, $R^1$ and $R^3$ are same as described above) is carried out in the manner described below.

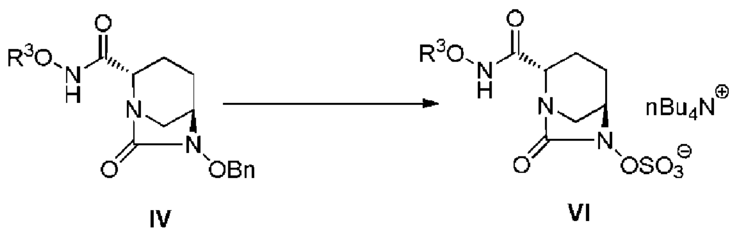
**[0054]** Examples of solvent used include water, methanol, ethanol, isopropanol, ethyl acetate, tetrahydrofuran, dioxane, dichloromethane, chloroform, 1,2-dichloroethane and 2,2,2-trifluoroethanol, preferable examples include ethyl acetate, dioxane, dichloromethane, chloroform and dichloroethane, and these solvents are used alone or as a mixture.

**[0055]** The compound $R^3ONH_2$ used in the reaction is selected from those listed as specific examples of $R^3$ and is used within the range of 1-2 equivalents, and preferably 1-1.3 equivalents, based on the compound of formula (III).

**[0056]** Examples of base used in the reaction include sodium hydrogencarbonate, potassium hydrogencarbonate, sodium carbonate, potassium carbonate, cesium carbonate, sodium hydroxide, potassium hydroxide, triethylamine, diisopropylethylamine, dimethylbutylamine, tributylamine, N-methylmorpholine, pyridine, N-methylimidazole, 4-dimethylaminopyridine, and triethylamine is used preferably, and the base can be used in the form of an aqueous solution in the case of using an inorganic base. The base is used within the range of 0-2 equivalents, and preferably 0-1.5 equivalents, based on the compound of formula (III). The reaction temperature is -25 to 50°C and preferably -10 to 10°C. The reaction time is 1-24 hours and preferably 1-16 hours.

**[0057]** The compound of formula (IV) can be isolated after completion of the reaction by diluting the reaction solution with a suitable solvent and sequentially washing with water, diluted acid and an aqueous basic solution (such as dilute hydrochloric acid, potassium hydrogensulfate, citric acid and aqueous sodium bicarbonate or saturated brine) followed by concentrating by evaporating the solvent. Examples of organic solvents used for dilution include diethyl ether, ethyl acetate, butyl acetate, toluene, dichloromethane and chloroform, and ethyl acetate is preferable. Although the product is isolated by ordinary work-up and purification procedures, it can be used in the next step after work-up only.

**[0058]** The step for converting a compound of Formula (IV):



to a compound of formula (VI) above is carried out in the manner described below.

**[0059]** Examples of solvents used in the reaction include water, methanol, ethanol, isopropanol and acetonitrile, and these solvents are used alone or as a mixture. In the case of carrying out the debenzylation reaction and sulfation reaction simultaneously, a hydrous solvent is preferable, and in the case of carrying out continuously, water is preferably added during sulfation. The amount of water added is 50-200 vol.-%, and preferably 75-125 vol.-%, of the solvent.

**[0060]** The amount of palladium carbon used is 5-100 wt.-%, and preferably 5-30 wt.-%, based on the compound of formula (IV).

**[0061]** The supply source of hydrogen used in hydrogenolysis is hydrogen gas, and the hydrogen pressure is selected within a range of atmospheric pressure to 1 MPa and preferably from atmospheric pressure to 0.5 MPa. The amount of hydrogen supplied is at least that used in an amount equal to or greater than the stoichiometric amount.

**[0062]** The reaction temperature of hydrogenolysis is 10-50°C and preferably 20-30°C. The reaction time is 0.5-3 hours and preferably 0.5-2 hours.

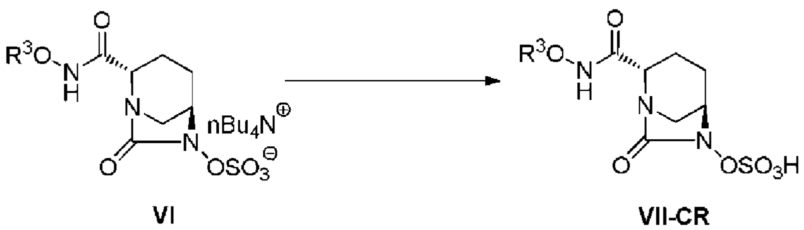
**[0063]** The sulfur trioxide-trimethylamine complex used as a sulfation reagent is used within the range of 1-2 equivalents, and preferably 1-1.3 equivalents, based on the compound of formula (IV).

**[0064]** Examples of base used in sulfation include triethylamine, tributylamine, diisopropylethylamine, N-methylmorpholine and disodium hydrogenphosphate, and triethylamine is used preferably, and the base can be used within the range of 0.1-1 equivalent, and preferably 0.1-0.3 equivalents, based on the compound of formula (IV).

**[0065]** The reaction temperature of sulfation is 0-50°C and preferably 15-30°C. The reaction time is 12-48 hours and preferably 12-24 hours.

**[0066]** After completion of the reaction, the compound of formula (VI) can be obtained by filtering out impurities such as catalyst and adding a concentrate obtained by solvent evaporation to an aqueous sodium dihydrogenphosphate solution, followed by adding 1-3 molar equivalents of tetrabutylammonium hydrogen sulfate, extracting with an organic solvent such as ethyl acetate and concentrating by evaporating the solvent. The resulting compound of formula (VI) can be used as a concentrated solution to the next step without isolating or purifying.

**[0067]** The step for converting a compound of Formula (VI):



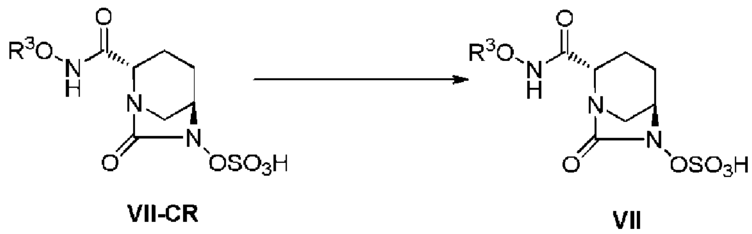
to a compound of formula (VII-CR) above is carried out in the manner described below.

**[0068]** Examples of solvents used in the step for deprotecting a protecting group optionally contained in $R^3$O-NHC(=O)-, particularly a tert-butoxycarbonyl (Boc) group, include ethyl acetate, tetrahydrofuran, dioxane, dichloromethane, chloroform, 1,2-dichloroethane and 2,2,2-trifluoroethanol, and dichloromethane or ethyl acetate is used preferably, and the solvent is used alone or as a mixture. The amount of solvent used is 2-10 vol/wt, and preferably 2-6 vol/wt, based on the net weight of the compound of formula (VI).

**[0069]** Examples of acids used in deprotection include hydrochloric acid, sulfuric acid, phosphoric acid, formic acid, trifluoroacetic acid, methanesulfonic acid, trifluoromethanesulfonic acid, chloromethanesulfonic acid and tetrafluoroboric acid, and trifluoroacetic acid is preferable. The acid is used within the range of 2-10 vol/wt, and preferably 2-6 vol/wt based on the net weight of the compound of formula (VI). The acid is added at -50 to 0°C and preferably -20 to 0°C. The reaction temperature is -5 to 20°C and preferably -5 to 5°C. The reaction time is 0.5-5 hours and preferably 0.5-3 hours.

**[0070]** Following completion of deprotection, the step for obtaining a compound of formula (VII-CR) by cooling the reaction solution and adding a poor solvent is carried out in the manner described below. Examples of the poor solvent used include ether-based poor solvents and ester-based poor solvents, preferable examples include ester-based poor solvents such as ethyl acetate, isopropyl acetate or butyl acetate, and ethyl acetate is even more preferable. The amount of poor solvent used is 1-3 times, and preferably 1.5-2 times, the volume of the reaction solution. The poor solvent, preliminarily cooled to -5 to 5 °C, is added to the reaction solution by dividing or dropping. Following completion of addition of the poor solvent, the reaction solution is additionally stirred. Precipitated material is filtered and washed several times as necessary, the resulting wet solid is subjected to vacuum drying and dried for 10 hours or more until the temperature of the material reaches room temperature to obtain the compound of formula (VII-CR).

**[0071]** The step for converting a compound of Formula (VII-CR):



to a compound of formula (VII) above is carried out in the manner described below.

**[0072]** A phosphate buffer having a pH of 6.5 is preferable for the buffer used to adjust the pH of the compound of formula (VII-CR). The concentration of the buffer is 0.1-0.5 M and is preferably 0.2 M, although varying according to remaining of the trifluoroacetic acid in the compound of formula (VII-CR). The total amount of buffer used is 10-50 vol/wt based on the net weight of the compound of formula (VII-CR).

**[0073]** Following 5-7 vol/wt of phosphate buffer is preliminarily cooled to 0-10°C, the pH is adjusted by alternately adding and dissolving small part of the compound of formula (VII-CR) and the cooled phosphate buffer so that the pH is 4-5.5 and preferably 4.2-4.8 and ultimately adjusting to pH 4.6, followed by diluting by adding water to 25 vol/wt in the case the total amount is less than 25 vol/wt based on the net weight of the compound of formula (VII-CR), and concentrating under reduced pressure to 20 vol/wt at a solution temperature of ≤ 15°C. Moreover, the pH of the aqueous solution is adjusted to 5.4 with phosphate buffer followed by diluting with water to obtain an aqueous solution having a concentration of 36 vol/wt.

**[0074]** Desalting of the aqueous solution following the aforementioned adjustment of pH is carried out as necessary by column purification using a synthetic adsorbent. Examples of synthetic adsorbents used include Diaion HP-20 and SepaBeads SP-207, Mitsubishi Chemical, and SepaBeads SP-207 is used preferably. The synthetic adsorbent is used within a range of 55 to 65 vol/wt based on the net weight of the compound of formula (VII-CR). Desalting is carried out by adsorbing the aforementioned aqueous solution to the synthetic adsorbent, desalting with 65-70 vol/wt of water, and eluting with 150-200 vol/wt of 10% isopropanol. The active fraction is contained at 20-25 vol/wt. Crystallization is carried out using a concentrate obtained by concentrating the resulting active fraction under reduced pressure to 5-7 vol/wt at a solution temperature of ≤ 20°C.

**[0075]** Crystal polymorphs may be solvated or anhydrous (such as an anhydride, mono hydrate or dihydrate). Crystalline forms I, II, III and IV exist as crystal polymorphs of a compound of formula (VII-1). Crystalline forms I and III crystallize at 0-35°C and preferably room temperature of 20-25 °C, more preferably as a result of seeding with a seed crystal, while crystalline form II can crystallize in the absence of seed crystal under supersaturated conditions by cooling at ≤ 15 °C and addition of poor solvent. Crystalline form IV can crystallize at room temperature using alcohol-based solvents such as methanol which dissolve a compound of formula (VII-1) well as poor solvent or be obtained by recrystallizing crystalline forms I to III in an alcohol solvent under a dehydrating condition.

**[0076]** Crystalline forms I, II and III are distinguished by DSC, solubility in aqueous isopropanol, and lattice spacing in powder X-ray diffraction patterns. Crystalline form I has the lowest solubility in aqueous isopropanol, while crystalline forms II and III have similar solubility.

**[0077]** Crystallization of the compound of formula (VII) is carried out in the manner described below. The initial amount of chemical solution is adjusted to a concentration that allows the crystalline form having the lowest solubility to adequately dissolve. In the case of the compound of formula (VII-1), the initial amount is 10-30% and preferably 10-20%. In the case of a crystalline form for which seeding is preferred, seed crystal is prepared in advance. Seed crystals of crystalline form I are acquired at room temperature, while seed crystals of crystalline form III can be obtained in free of contamination by successive crystallization within the range of 15-25%. The amount of seed crystal used is 0.01-20% and preferably 0.01-2%.

**[0078]** Examples of poor solvents used include methanol, ethanol, isopropanol, acetone, acetonitrile and tetrahydrofuran, and preferably include alcohol-based poor solvents such as methanol, ethanol or isopropanol. The amount of the poor solvent is adjusted based on solubility so that isolation loss is ≤ 1%. In the case of the compound of formula (VII-1), the poor solvent is used at 1-10 times, and preferably 6-7 times, the initial volume of the solution. The timing of the addition of poor solvent is such that the poor solvent is dropped in after the mixture has formed a slurry following seeding in the case of crystalline form I, immediately after seeding in the case of the crystalline forms II and III, and without seeding in the case of the crystalline form IV.

**[0079]** The crystalline form IV can be also obtained by suspending and stirring the crystalline form I, II or III in methanol, ethanol or isopropanol instead of making an aqueous solution thereof.

**[0080]** Controlling the temperature of the solution is an important factor in terms of controlling the desired crystal polymorphism and is determined by referring to the precipitation rate of crystal polymorphs at a set temperature. In the case of a compound of formula (VII-1), crystalline forms I, III and IV are controlled to 20-25°C, while crystalline form II